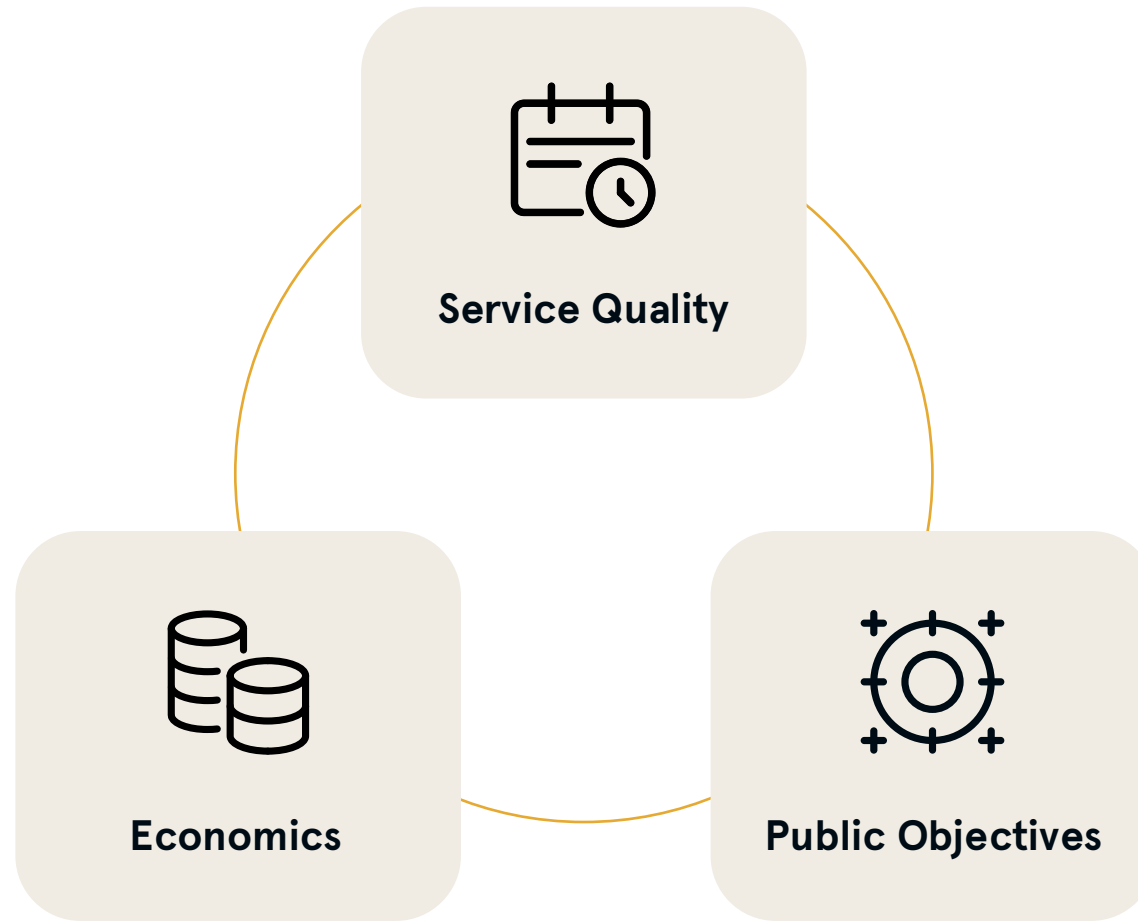


MATSim in Production

**Live Demand Prediction, Automated DRT Scenario
Build and Visualization for Non-Expert Users**

Tom-Erik Kuhlen, MOIA GmbH

We help cities and mobility companies to plan an AD-service



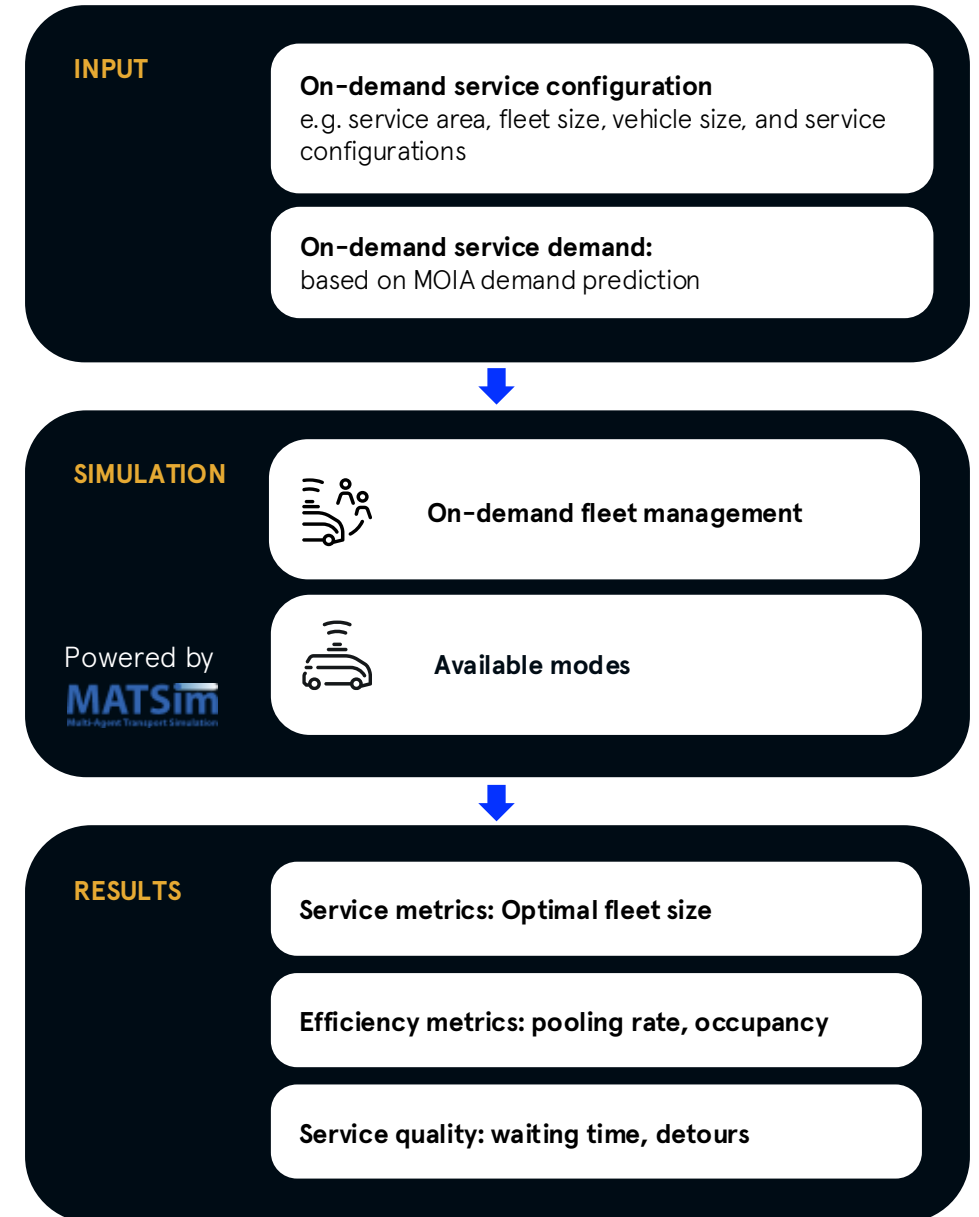
Fleet Simulation

Fastest way to get insights into the potential of autonomous on-demand services in your region

Answers questions like:

- How many vehicles are required?
- What is the average waiting time?
- What is the operational profit?

Best for: Scenario evaluation

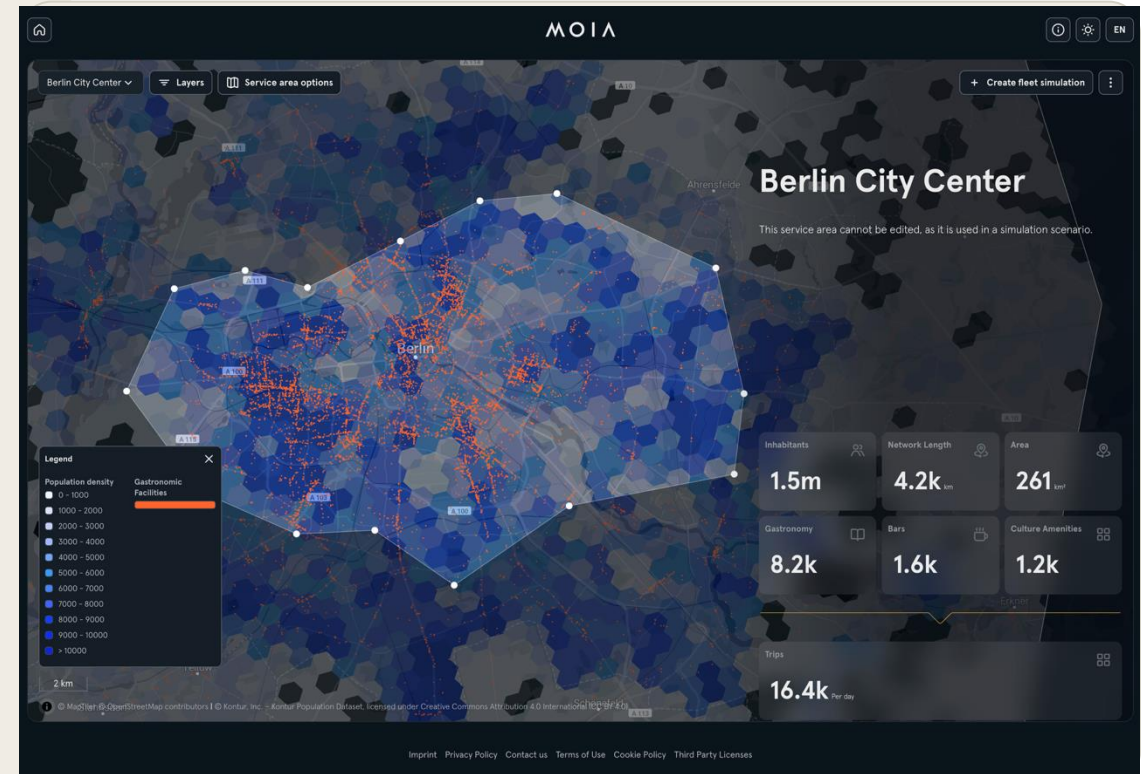


Live MIA Session

MIA Explorer: Find the optimal service area based on MOIA Demand Prediction

- Layers activated: population density and gastronomy
- Service area drawn based on data layers
- Predicted demand appears live for this area
- Population and points of interest as inputs

Note: This is predicted demand based on MOIA's Demand Prediction, not current ridership, so it also works for regions that have no service today



MIA Explorer, Berlin city center service area with population density and gastronomic facility layer activated

One region, two service promises

Same service area, same predicted demand. Only the promise to the passenger changes.

Basic service

cheap to run, asks for patience from passengers

Max waiting time: 30 min

Max detour: 60 %

Premium service

a quality promise to the passenger

Max waiting time: 10 min

Max detour: 20 %

What a service promise costs

The questions from earlier

- How many vehicles are required?
- What is the average waiting time?
- What is the operational profit?

	Basic service	Premium service	Difference
Vehicles needed	129	287	+122 %
Average waiting time achieved	19 min	5 min	- 74 %
Total kilometers traveled	58k km	72k km	+ 24 %
Operational Profit	123k €	98k €	- 20 %

Quality costs vehicles, and tighter time windows lower the efficiency and therefore increase the total kilometers traveled. Both effects push cost per ride in the same direction, which lowers the profit. The trade-off is now a decision based on numbers.

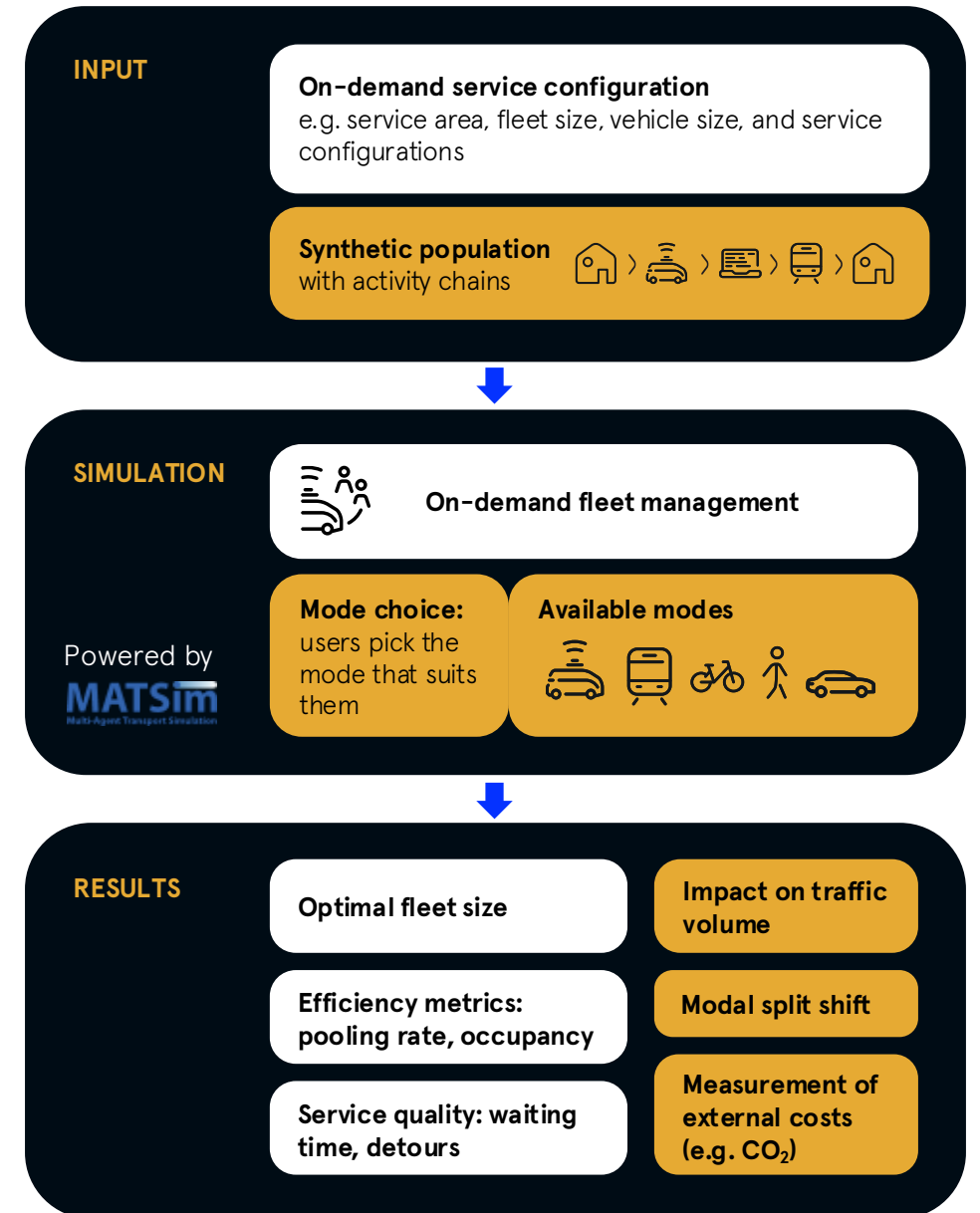
Soon: Mobility Simulation in MIA

Simulate your region's mobility system end to end and align every stakeholder

Answers questions like:

- How does the modal split change?
- What is the optimal pricing model?
- What is the effect on sustainability?

Best for: Policy assessments + system-wide analysis



Try MIA – We would love to get your feedback!



Scenario Creation

what additional parameters would you expect?



Presentation

which additional KPIs are missing from the dashboard, and are the current ones understandable?



Guardrails

which parameter combinations would you warn about?



Request a free trial

1. Create your MIA user on mia.vwgroup.cloud
2. Send us a mail to request a free trial

Q&A

MOIA

Let's simulate your region to discover the true potential of autonomous on-demand services!



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